



PC-based multimeter (Courtesy: National Instruments)

electronics have improved the measurement accuracy. Older analogue meters had basic accuracies of 5 to 10 per cent. On the other hand, today's portable DMMs have accuracies as good as ± 0.005 per cent, while bench-top instruments have accuracies in single-digit parts per million figures."

DMM itself being an electronic device, it is necessary that its own electronics does not interfere with the measurements. "All digital meters contain a battery to power the display. So they use virtually no power from the circuit under test. This means that on their DC voltage ranges, they have a very high resistance (input impedance) of about 10 mega-ohms, and are very unlikely to affect the circuit under test," explains Joshi.

The design of the electronic circuitry has well adopted the use of every advanced piece of silicon available in the market. "Faster and more capable microprocessors help increase data transfer rates. Communication chips that handle USB and Ethernet are common. Scalable FPGAs are used along with ASICs to enhance both analogue and digital processing. FPGAs and ASICs allow for more measurements and more computations. Some instruments have USB ports for USB memory sticks, so data can be captured and transferred to any computer," says Green.

Some of the design considerations include the type of measurement and readings being displayed. Manjunath says, "Conventional DMMs that do not follow the true-RMS measurement are accurate as long as the AC signal is completely sinusoidal in shape. But today, due to increased use of electronic control equipment, we hardly find

any sinusoidal waveform in industrial and commercial environments. This highly inaccurate way of measurement may lead to failure in design, operation and maintenance of electrical equipment and cause severe damage to equipment and personal safety. So a true-RMS multimeter is a must."

"Use of surface-mount technology has made the design of equipment much more compact, reducing the size and weight of the equipment, while increasing the reliability of equipment considerably," adds Waghchoure.

High accuracy, reliability and safety

When it comes to measurement of electrical quantities like voltage and current, there is always some risk associated due to mishandling or harsh environmental conditions. Siddique points out that many a time users need to use the equipment in high-voltage and high-current environments, which may be harmful for the instrument as well as the user.

"Reliability and safety, especially under tough conditions, are more important than ever before," says Thakkar. He adds, "DMMs are designed with latest safety standards like the IEC category and voltage rating approved for the environment where the measurement is to be made. Choosing a meter with this rating which also has a UL, CSA, VDE or TÜV certification means the meter not only has been designed to IEC standards but has been independently tested to meet these standards."

"Accuracy and resolution usually found only in bench-top DMMs of the past are now common in their handheld counterparts," claims Thakkar. "Some equipment have basic DC voltage accuracy of up to 0.09 per cent and resolution of 6000 counts." All the meters can be calibrated and have traceability to industry standards, just like their bench counterparts.

Manjunath adds, "Overload protection prevents damage to both the meter

and the circuit while protecting the user. Also, some special high-energy fuses provide extra protection for the user and meter during current measurements and overloads."

Need of the Indian market

Multimeters is a highly competitive market where several established local and global players have created a market for themselves in India.

Siddique says, "The Asia Pacific region is a very important market for multimeters with maximum demand coming from China and India. End users in these regions are highly cost-conscious demanding the best price-value combination."

Vendors are offering unique features catering to various segments of the Indian market. Sai Venkat says, "With the requirements of flexible configurations, quick setup and affordability, the instruments have gained interest from the electronics industry, educators and students who continue to look for value-added instruments to achieve cost advantages in the ever-challenging world."

Apart from affordability, ease-of-use, safety and reliable testing, there is also a need for communication in local language. Today, vendors offer a lot of new features and capabilities in handheld multimeters, such as visual and audible continuity indication, IP-54 certification (for water- and dust-resistance), PC connectivity and CAT III/IV safety protection along with user manuals in diverse Indian languages such as Hindi, Bengali, Tamil, Malayalam and Kannada.

Thakkar says, "When evaluating the suitability of several digital multimeters, the best approach is to choose a set of measurements and conditions that approximates your application. Choosing a DMM for the job requires not only looking at basic specifications but also the features, functions and the overall value represented by a meter's design and the care taken in its production." ●

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